ABDULLAH RIAZ

GIS Developer | Remote Sensing Researcher | Geospatial Applications

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PROFESSIONAL PROFILE

GIS Developer & Remote Sensing Researcher with a BS in GIS and an MPhil in Remote Sensing & GIS. Strong expertise in Web GIS development, spatial databases, Earth Observation, and Google Earth Engine (GEE). Proven track record in applying machine learning to multi-factor satellite data, including glacier surge detection, landslide susceptibility modeling, and groundwater quality analysis. Experienced in building public-sector geospatial dashboards, seeking to leverage GeoAI and cloud-based Earth Observation to address complex cryosphere and environmental hazards.

EDUCATION

Sep 2025 - Present **M.Phil. in Geographic Information System and Remote Sensing (GIS & RS)**

Centre for Geographical Information System, University of the Punjab | Lahore, Pakistan

- CGPA: 3.80/4.00
- Thesis: Machine learning-based detection of glacier surges in the Shigar basin, Karakoram region using multi-sensor remote sensing data.

Sep 2021 - Jun 2025 **BS in Geographic Information System (GIS)**

Centre for Geographical Information System, University of the Punjab | Lahore, Pakistan

- CGPA: 3.70/4.00
- Thesis: A comparative analysis of groundwater quality using GIS based technique for WASA administrative towns of Lahore City.

Aug 2019 - Jul 2021 **FSc Pre-Engineering**

Govt. Graduate College Township | Lahore, Pakistan

- Grade: A+

Mar 2017 - Apr 2019 **Matriculation in Science**

Mubarik Educare System | Lahore, Pakistan

- Grade: A+

WORK EXPERIENCE

Aug 2025 - Present **GIS Developer**

THE SPATIO, Engineering & Geospatial Consultants | Lahore, Pakistan

- Developed frontend and interactive mapping modules for the PWASA Smart Utility Dashboard.
- Designed and developed WebGIS components for the Suthra Punjab Microplan Dashboard with full-stack integration.
- Contributed to a public parcel-display dashboard for the PULSE project.
- Worked with administrative-boundary data, spatial filtering workflows, dashboard visualization, and spatial database-supported applications.

Apr 2025 - Jul 2025 **Machine Learning Intern**

Punjab Information Technology Board (PITB) | Lahore, Pakistan

- Applied machine-learning methods to crop classification in Punjab using Sentinel-2 imagery.
- Conducted spatial analysis of pilgrimage zones using ArcGIS and Google Earth Engine.
- Processed and analyzed drone imagery for environmental monitoring and assessment.

May 2023 - May 2025 **Teaching Assistant**

Centre for Geographical Information System, University of the Punjab | Lahore, Pakistan

- Assisted more than 50 undergraduate students in Java and OOP concepts.
- Provided practical support using NetBeans and helped students debug programming assignments.

Jul 2024 - Aug 2024 **GIS Analyst Intern**

World Wide Fund for Nature (WWF) Pakistan | Lahore, Pakistan

- Used Google Earth Engine and QGIS for environmental and ecological spatial analysis.
- Processed and interpreted geospatial datasets for conservation-related assignments.
- Supported mapping and visualization of environmental information for project activities.

PROJECTS

Suthra Punjab GIS Dashboard: Developed Web GIS interfaces and spatial filtering workflows for monitoring solid-waste operations across administrative units.

PWASA Smart Utility Dashboard: Developed frontend and interactive mapping components for visualization and management of utility-related spatial information.

PULSE Public Parcel Display Platform: Supported the development of a dashboard for public visualization of land-parcel information.

Sentinel-2 Crop Classification: Applied machine-learning methods to satellite imagery for agricultural crop classification in Punjab.

Glacial Lake Outburst Flood (GLOF) Risk Assessment: Analyzed the 2019 GLOF event in Golen Gol, Chitral, using geospatial and field data to map high-risk zones and assess landslide vulnerability through machine learning models.

Vegetation Stress Hotspot Calculation: Pre-processed high resolution drone imagery and applied indices (NDVI, NDWI, SAVI, and MSAVI) to detect crop stress and assess vegetation health.

Hypsometric & Soil Loss Erosion Analysis: Performed hypsometric & soil erosion analysis of the Naulong Watershed (Jhal Magsi) Balochistan using RUSLE model to assess erosion potential and watershed stability.

Assessment of Groundwater Quality: Evaluated decadal groundwater quality trends in Gulberg & Nishtar towns using GIS based Fuzzy logic & Entropy models, and created spatial water quality index (WQI) maps to identify contamination hotspots.

TECHNICAL SKILLS

Programming	Python, JavaScript, SQL, Java
Web GIS and Development	React, frontend development, Web GIS applications, interactive dashboards, geospatial data visualization
Spatial Databases	PostgreSQL/PostGIS, Oracle
Earth Observation and Remote Sensing	Google Earth Engine, Sentinel-2 processing, image classification, change detection, drone image processing
GIS Software	ArcGIS Pro, QGIS, ENVI, ESA SNAP
Research and Analysis	Spatial modelling, environmental assessment, research methodology, scientific writing, and accuracy assessment
Development Tools	Git and GitHub
Languages	Urdu - Native; English - Professional working proficiency; Turkish – Proficient; German – Intermediate

COMMUNICATION, TEAMWORK AND PROFESSIONAL SKILLS

- Explained Java and object-oriented programming concepts to more than 50 undergraduate students as a Teaching Assistant.
- Collaborated with GIS analysts, developers, researchers, and public-sector stakeholders on geospatial dashboard projects.
- Communicated research findings through peer-reviewed publications, academic presentations, and technical project discussions.
- Managed academic research, teaching responsibilities, and professional GIS assignments concurrently while meeting project deadlines.
- Worked effectively in academic, governmental, private-sector, and non-governmental environments.

RESEARCH PUBLICATIONS

- **Riaz, A.,** Usman, S. M., Chaudhary, M. H., Samad, A., Gulzar, Q., & Ilyas, M. (2025). A GIS-Based Comparative Analysis of Ground Water Quality in Administrative Towns of Lahore City (2014-2024). International Journal of Innovations in Science & Technology, 7(9), 114.
- Iqbal, U., Shah, M. H., & **Riaz, A.** (2025). Geo-Spatial Analysis to Access Land Slide Susceptibility in Tehsil Balakot, District Mansehra, Pakistan. International Journal of Innovations in Science & Technology, 7(3), 2104-2116. DOI: 10.33411/ijist/20257321042116.